Economic Resilience in OECD Countries: A Structural Data-Driven Framework

1. Objective and Research Definition

The goal of this project is to construct a **quantitative and interpretable framework for economic resilience across OECD countries**, shifting the focus from traditional performance metrics (e.g., GDP growth) to **structural economic stability under systemic shocks**.

Research Question

> Which structural socio-economic characteristics are associated with higher economic resilience in OECD countries during systemic shocks?

Core Idea

Economic resilience is not defined as economic growth or performance in stable conditions, but as:

> **the ability of an economic system to absorb external shocks while maintaining structural stability and recovering key macroeconomic indicators.**

This implies a shift from:

- Output maximization → to stability under stress
- Short-term growth → to long-term adaptive capacity

---

2. Data Sources (OECD Database)

All primary data will be extracted from the OECD Data Explorer platform.

Main datasets include:

- Macroeconomic indicators:
  - GDP per capita (level and growth)
  - Inflation rate
  - Unemployment rate
  - Productivity measures
- Structural economic indicators:
  - R&D expenditure (% GDP)
  - Patent applications / innovation indicators
  - Education attainment levels
  - Labour market participation
- Institutional & social indicators:
  - Institutional trust
  - Income inequality (Gini coefficient)
  - Social support indicators
  - Public debt levels
- Energy & structural dependency:
  - Energy import dependency
  - Housing cost burden
  - Sectoral concentration proxies

Time dimension

Panel data across multiple years (pre- and post-shock periods, e.g. 2000–2024 depending on availability).

---

3. Construction of the Economic Resilience Framework (ERF)

The central output of the project is not a single scalar resilience index, but a **structural comparison framework** designed to analyse how different economic systems behave under multiple resilience-relevant dimensions.

The methodology is based on **Partial Order Theory (POSets)** and Pareto dominance relations, avoiding artificial aggregation into a single composite score.

This approach is motivated by the assumption that economic resilience is:

- multi-dimensional,
- structurally heterogeneous,
- and not fully reducible to a one-dimensional metric.

Step 1 – Selection of OECD Variables

The framework starts from directly observable OECD socio-economic variables, selected according to their theoretical relevance to structural resilience, found in point 2.

The objective is not to estimate "resilience" directly, but to compare countries through measurable structural characteristics associated with adaptive capacity under systemic stress.

Step 2 – Data Preprocessing

Before comparison, the dataset undergoes standard preprocessing procedures:

- Missing value handling
- Temporal alignment across countries and years
- Removal of duplicated or inconsistent observations
- Directional alignment of variables
- Statistical normalization

Directional alignment ensures that:

- higher values always correspond to structurally preferable conditions.

For example:

- unemployment and debt exposure are inversed,
- innovation and education remain positively oriented.

Normalization is required because variables operate on different scales and units.

Step 3 – Partial Order Construction

Instead of aggregating variables into intermediate latent indexes, countries are compared directly in the multidimensional variable space.

A country A is considered structurally dominant over country B if:

- A is not worse than B across all considered variables,
- and strictly better in at least one dimension.

Formally: $A \succeq B$

This produces a **partial ordering** of OECD countries rather than a forced global ranking.

The resulting structure allows the identification of:

- structurally dominant economies,
- dominated economies,
- and incomparable systems characterized by different trade-offs.

This is particularly important because two countries may exhibit:

- similar economic performance,
- but fundamentally different structural vulnerabilities and adaptive capacities.

Step 4 – Pareto Structure and Structural Interpretation

The POSets framework naturally generates:

- Pareto frontiers,
- dominance hierarchies,
- and clusters of structurally similar economies.

Countries positioned near the Pareto frontier are interpreted as systems exhibiting stronger combinations of resilience-relevant characteristics.

Importantly, this does not imply "optimality" in an absolute sense, but relative structural robustness under the selected dimensions.

The methodology therefore avoids:

- arbitrary weighting schemes,
- excessive dimensional reduction,
- and artificial precision associated with traditional composite indicators.

Step 5 – Shock-Based Validation of Structural Robustness

The validity of the ERF is assessed empirically by testing how structural positions relate to observed resilience during systemic shocks (e.g. 2008 financial crisis, COVID-19, energy crisis/ inflation shock (2021–2023)).

Countries identified as:

- Pareto-dominant or structurally central

  are expected to show:

- lower GDP and employment volatility
- faster recovery trajectories
- higher stability across macroeconomic indicators

This validation step does not assume prediction of future shocks, but evaluates **consistency between structural configuration and observed adaptive performance under past stress events**.

Outcome of the Framework

The ERF does not produce a single "resilience score", but instead provides:

1. A **structural map of OECD economies**
2. A **partial ordering of resilience profiles**
3. Identification of **different types of resilient configurations**
4. An empirical link between **economic structure and shock response behaviour**

This allows the analysis to move from a reductive ranking approach to a **system-level comparison of economic architectures under uncertainty**.

---

4. Expected Output

The project will produce:

1. **Economic Resilience Index (ERI)** for OECD countries
2. **Country clustering based on structural resilience profiles**
3. **Empirical validation of which structural factors matter most**

4. **Policy and business implications for investment and risk assessment**

5. Limitations

- Resilience is not directly observable and is approximated through proxies
- Shock effects may vary across countries and sectors
- Causality cannot be fully established (only structural association)
- Data availability may differ across OECD indicators and years

6. Conclusion

This project proposes a shift from traditional macroeconomic evaluation toward a **structural resilience framework**. Instead of focusing on growth or performance in isolation, it aims to understand which underlying economic structures enable countries to remain stable under systemic stress.

The final output is not a predictive model of the future, but a **decision-support framework for assessing systemic economic robustness**, applicable to policy analysis, strategic investment, and macroeconomic risk evaluation.